

School Math Workbook Targeted Mastery Worksheet (Set 12)

Module: Number Theory · Prime Factorization · Fractions Operations & 3D Geometry

Name: _____

Class: _____

Date: _____

Total: 100
pts

Score: _____

Section 1: Factors, Multiples & Primes

2 pts each, 20 pts total

1. List all positive factors in ascending order without omissions:

(1) All positive factors of 28: _____ ;

(2) All positive factors of 48: _____ .

2. Among the numbers 77, 83, 91, 97, the prime numbers are _____ , and the composite numbers are _____ .

3. Understand the meaning of powers and evaluate:

$$2^3 = \underline{\quad\quad} \quad 2^4 = \underline{\quad\quad} \quad 3^4 = \underline{\quad\quad} \quad 10^0 = \underline{\quad\quad} \quad 10^4 \times 10^5 = 10^{\underline{\quad\quad}}$$

4. The area of a square with an integer side length is called a square number. Among the integers from 1 to 20, the square numbers are _____ ; the number 6 is _____ a square number (write "is" or "is not").

5. Determine whether the outcome of each expression is Odd or Even:

(1) 3^2 is an _____ number (2) 6^2 is an _____ number (3) If n is even, $n^2 + 1$ is an _____ number

6. Consider the 5-digit number $\overline{4271\square}$: if it is divisible by 2, the digit $\square$ can be _____ ; if it is divisible by 3, the digit $\square$ can be _____ ; if it is divisible by both 2 and 5, the digit $\square$ must be _____ .

7. Express the prime factorization of each number in Index Form:

$$81 = \underline{\quad\quad\quad} \quad 72 = \underline{\quad\quad\quad}$$

8. Given two positive integers $E = 2^{10} \times 3^4$ and $F = 2^6 \times 3 \times 5^5$. In index form, their Highest Common Factor (HCF) is _____ ; their Lowest Common Multiple (LCM) is _____ .

9. The HCF of two positive integers a and b is 6, and their LCM is 90. If $a = 18$, then $b = \underline{\quad\quad\quad}$.

10. For $52 \div 24$, the quotient written in division-with-remainder form is _____ , and written as a mixed number in its simplest form is _____ .

Section 2: Prime Factorization & HCF/LCM

20 pts total

1. Use short division (divisors must be prime numbers) to perform prime factorization, and write out the final prime product (4 pts each, 8 pts total):

(1) Prime factorization of 54

[54 = _____]

(2) Prime factorization of 1080

[1080 = _____]

2. Use short division to find the HCF and LCM for each set of numbers (4 pts each, 8 pts total):

(1) Find HCF and LCM of 54 and 90

[HCF = _____, LCM = _____]

(2) Find HCF and LCM of 24, 36 and 48

[HCF = _____, LCM = _____]

3. Counting Factors Investigation (4 pts):

The prime factorization of 72 is $72 = 2^3 \times 3^2$. Use the prime exponent law to calculate how many positive factors 72 has in total. Show the calculation expression.

[Expression: _____ , Total factors: _____]

Section 3: Fractions Operations

20 pts total

1. Mental calculations and conversions (1 pt each, 4 pts total):

(1) $\frac{23}{6} = \underline{\hspace{2cm}}$
(mixed)

(2) $2\frac{3}{7} = \underline{\hspace{2cm}}$
(improper)

(3) $\frac{3}{6} + \frac{5}{6} = \underline{\hspace{2cm}}$
(simplest mixed)

(4) $1 - \frac{5}{12} = \underline{\hspace{2cm}}$

2. Addition and subtraction of unlike fractions (show common denominators, 2 pts each, 8 pts total):

① $\frac{2}{3} + \frac{1}{5}$

[Ans: _____]

② $2\frac{1}{3} + 3\frac{5}{6}$

[Ans: _____]

③ $4\frac{1}{4} - 1\frac{2}{3}$

[Ans: _____]

④ $\frac{5}{8} - \frac{1}{6} + \frac{7}{12}$

[Ans: _____]

3. Fraction multiplication and division (2 pts each, 8 pts total):

① $\frac{5}{12} \div \frac{15}{16}$

[Ans: _____]

② $\frac{7}{18} \times \frac{9}{14} \div \frac{5}{6}$

[Ans: _____]

③ $\left(\frac{3}{4} + \frac{1}{6}\right) \times 12$

[Ans: _____]

④ $\frac{5}{9} \div \left(\frac{2}{3} - \frac{1}{4}\right)$

[Ans: _____]

Section 4: 3D Spatial Geometry & Blocks

20 pts total

1. Fundamental Rectangular Prism Calculations (6 pts):

A rectangular prism has length 5 cm, width 4 cm, and height 3 cm.

- (1) Calculate the total surface area of this rectangular prism in cm^2 .
- (2) Calculate the volume of this rectangular prism in cm^3 .

[(1) Surface Area: _____ cm^2 ; (2) Volume: _____ cm^3]

2. Stacking and Joining Identical Prisms (6 pts):

There are two identical rectangular prisms, each measuring 5 cm by 3 cm by 2 cm. They are joined along one entire face to form a single larger solid.

- (1) What is the total surface area of the combined solid if the glued face has maximum area?
- (2) What is the total surface area of the combined solid if the glued face has minimum area?

[(1) Max glued face area solid: _____ cm^2 ; (2) Min glued face area solid: _____ cm^2]

3. Tiered Composite Block Model (8 pts):

Unit cubes with edge length 1 cm are arranged on a tabletop to build a 3-tier stepped solid: the bottom tier is a 3×3 grid (9 cubes); the middle tier is a 2×2 grid placed flush against one corner (4 cubes); the top tier consists of 1 cube flush against that same corner.

- (1) How many unit cubes are used in total? What is the total volume of the solid in cm^3 ?
- (2) All exposed surfaces (excluding the bottom resting on the table) are painted. What is the total painted surface area in cm^2 ?

[(1) Total cubes: _____ , Volume: _____ cm^3 ; (2) Painted Area: _____ cm^2]

Section 5: High-Order Problem Solving

5 pts each, 20 pts total

1. Divisibility Rules & Multiple Branches (5 pts):

The 6-digit number $\overline{5a342b}$ is divisible by 2, 3, and 5 simultaneously.

- (1) What is the value of the last digit b ?
- (2) Find all possible values of digit a .

[(1) $b =$ _____ ; (2) All possible values of a : _____]

2. Statistical Measures: Mean, Median, Mode & Range (5 pts):

A basketball player scored the following points in 10 consecutive games:

24, 28, 27, 31, 27, 35, 27, 29, 31, 41

- (1) Find the Range and the Mode of this dataset.
- (2) Find the Median and the Mean of this dataset.

[(1) Range: _____ , Mode: _____ ; (2) Median: _____ , Mean: _____]

3. Periodic Cycle & Starting Point (5 pts):

A playlist of 5 songs plays continuously in order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A \rightarrow \dots$

(1) If the playback starts from song A , which song is played 43rd?

(2) If song C is currently playing (counted as the 1st song), which song is played when the count reaches 38?

[(1) The 43rd song is _____ ; (2) The 38th song is _____]

4. Encounter Journey & Fraction Modeling (5 pts):

Person A and Person B start simultaneously from two points 1000 m apart and travel towards each other. Person A walks at 60 m/min, and Person B rides a bicycle at 140 m/min.

(1) After how many minutes will they meet?

(2) At the moment they meet, what fraction of the total distance has Person A traveled? How many more meters has Person B traveled than Person A?

[(1) Meet _____ min; (2) Person A _____ of total, Person B _____ m more after traveled traveled]

Teacher Reference Key & Full Step-by-Step Solutions (For Scoring & Guidance)

Section 1: Factors, Multiples & Primes

- (1) 1, 2, 4, 7, 14, 28; (2) 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.
- Primes: 83, 97; Composites: 77 (7×11), 91 (7×13).
- 8; 16; 81; 1; Exponent is 9 ($10^{4+5} = 10^9$).
- 1, 4, 9, 16; is not.
- (1) Odd; (2) Even; (3) Odd (square of even is even, plus 1 is odd).
- 0, 2, 4, 6, 8; 1, 4, 7 ($4 + 2 + 7 + 1 = 14$, adding 1, 4, or 7 yields a multiple of 3); 0.
- $81 = 3^4$; $72 = 2^3 \times 3^2$.
- HCF = $2^6 \times 3$; LCM = $2^{10} \times 3^4 \times 5^5$.
- $b = 30$ (using $a \times b = \text{HCF} \times \text{LCM} \implies 18 \times b = 6 \times 90 = 540 \implies b = 30$).
- 2 R 4; $2\frac{1}{6}$ ($2\frac{4}{24} = 2\frac{1}{6}$).

Section 2: Prime Factorization & HCF/LCM

- (1) Short division: $54 \div 2 = 27$, $27 \div 3 = 9$, $9 \div 3 = 3$. Product: $54 = 2 \times 3 \times 3 \times 3$ (or 2×3^3);
(2) Short division: $1080 \div 2 = 540$, $540 \div 2 = 270$, $270 \div 2 = 135$, $135 \div 3 = 45$, $45 \div 3 = 15$, $15 \div 3 = 5$. Product:
 $1080 = 2^3 \times 3^3 \times 5$.
- (1) 54 and 90: Common prime factors $2 \times 3 \times 3 = 18$ (HCF); quotient 3 and 5, LCM = $18 \times 3 \times 5 = 270$;
(2) 24, 36, 48: Common prime factors $2 \times 2 \times 3 = 12$ (HCF); LCM = 144.
- Formula: $(3 + 1) \times (2 + 1) = 4 \times 3 = 12$; 72 has 12 positive factors.

Section 3: Fractions Operations

- (1) $3\frac{5}{6}$; (2) $\frac{17}{7}$; (3) $1\frac{1}{3}$; (4) $\frac{7}{12}$.
- ① $\frac{2}{3} + \frac{1}{5} = \frac{10}{15} + \frac{3}{15} = \frac{13}{15}$;
② $2\frac{1}{3} + 3\frac{5}{6} = 2\frac{2}{6} + 3\frac{5}{6} = 5\frac{7}{6} = 6\frac{1}{6}$;
③ $4\frac{1}{4} - 1\frac{2}{3} = 4\frac{3}{12} - 1\frac{8}{12} = 3\frac{15}{12} - 1\frac{8}{12} = 2\frac{7}{12}$;
④ $\frac{5}{8} - \frac{1}{6} + \frac{7}{12} = \frac{15}{24} - \frac{4}{24} + \frac{14}{24} = \frac{25}{24} = 1\frac{1}{24}$.
- ① $\frac{5}{12} \div \frac{15}{16} = \frac{5}{12} \times \frac{16}{15} = \frac{4}{9}$;
② $\frac{7}{18} \times \frac{9}{14} \div \frac{5}{6} = \frac{1}{4} \times \frac{6}{5} = \frac{3}{10}$;
③ $(\frac{3}{4} + \frac{1}{6}) \times 12 = \frac{3}{4} \times 12 + \frac{1}{6} \times 12 = 9 + 2 = 11$;
④ $\frac{5}{9} \div (\frac{8}{12} - \frac{3}{12}) = \frac{5}{9} \div \frac{5}{12} = \frac{5}{9} \times \frac{12}{5} = \frac{4}{3} = 1\frac{1}{3}$.

Section 4: 3D Spatial Geometry & Blocks

- (1) Surface Area = $2 \times (5 \times 4 + 4 \times 3 + 5 \times 3) = 2 \times (20 + 12 + 15) = 94 \text{ cm}^2$;
(2) Volume = $5 \times 4 \times 3 = 60 \text{ cm}^3$.
- Single solid SA = $2 \times (15 + 6 + 10) = 62 \text{ cm}^2$. Sum of two = 124 cm^2 .
(1) Max glued face area ($5 \times 3 = 15 \text{ cm}^2$): New SA = $124 - 2 \times 15 = 94 \text{ cm}^2$;
(2) Min glued face area ($3 \times 2 = 6 \text{ cm}^2$): New SA = $124 - 2 \times 6 = 112 \text{ cm}^2$.
- (1) Total cubes = $9 + 4 + 1 = 14$; Volume = $14 \times 1^3 = 14 \text{ cm}^3$;
(2) Top projection area = $3 \times 3 = 9 \text{ cm}^2$. Each of the 4 side faces has area $3 + 2 + 1 = 6 \text{ cm}^2$, lateral area sum = $4 \times 6 = 24 \text{ cm}^2$. Total painted area = $9 + 24 = 33 \text{ cm}^2$.

Section 5: High-Order Problem Solving

- (1) Divisible by both 2 and 5 requires last digit $b = 0$;
(2) Sum of digits = $5 + a + 3 + 4 + 2 + 0 = 14 + a$. For $14 + a$ to be divisible by 3, a can be 1, 4, 7.
- (1) Range = $41 - 24 = 17$; Mode = 27 (appears 3 times);
(2) Ordered data: 24, 27, 27, 27, 28, 29, 31, 31, 35, 41. Median = $(28 + 29) \div 2 = 28.5$; Mean = $300 \div 10 = 30$.

3. (1) Period is 5. $43 \div 5 = 8 \text{ R } 3$, corresponding to the 3rd song: C ;

(2) From C, advance $38 - 1 = 37$ steps. $37 \div 5 = 7 \text{ R } 2$. Advance 2 songs from C ($C \rightarrow D \rightarrow E$): Song E .

4. (1) Time to meet = $1000 \div (60 + 140) = 1000 \div 200 = 5 \text{ min}$;

(2) Person A distance = $60 \times 5 = 300 \text{ m}$, fraction = $\frac{300}{1000} = \frac{3}{10}$; Person B distance = $140 \times 5 = 700 \text{ m}$, difference = $700 - 300 = 400 \text{ m}$.
